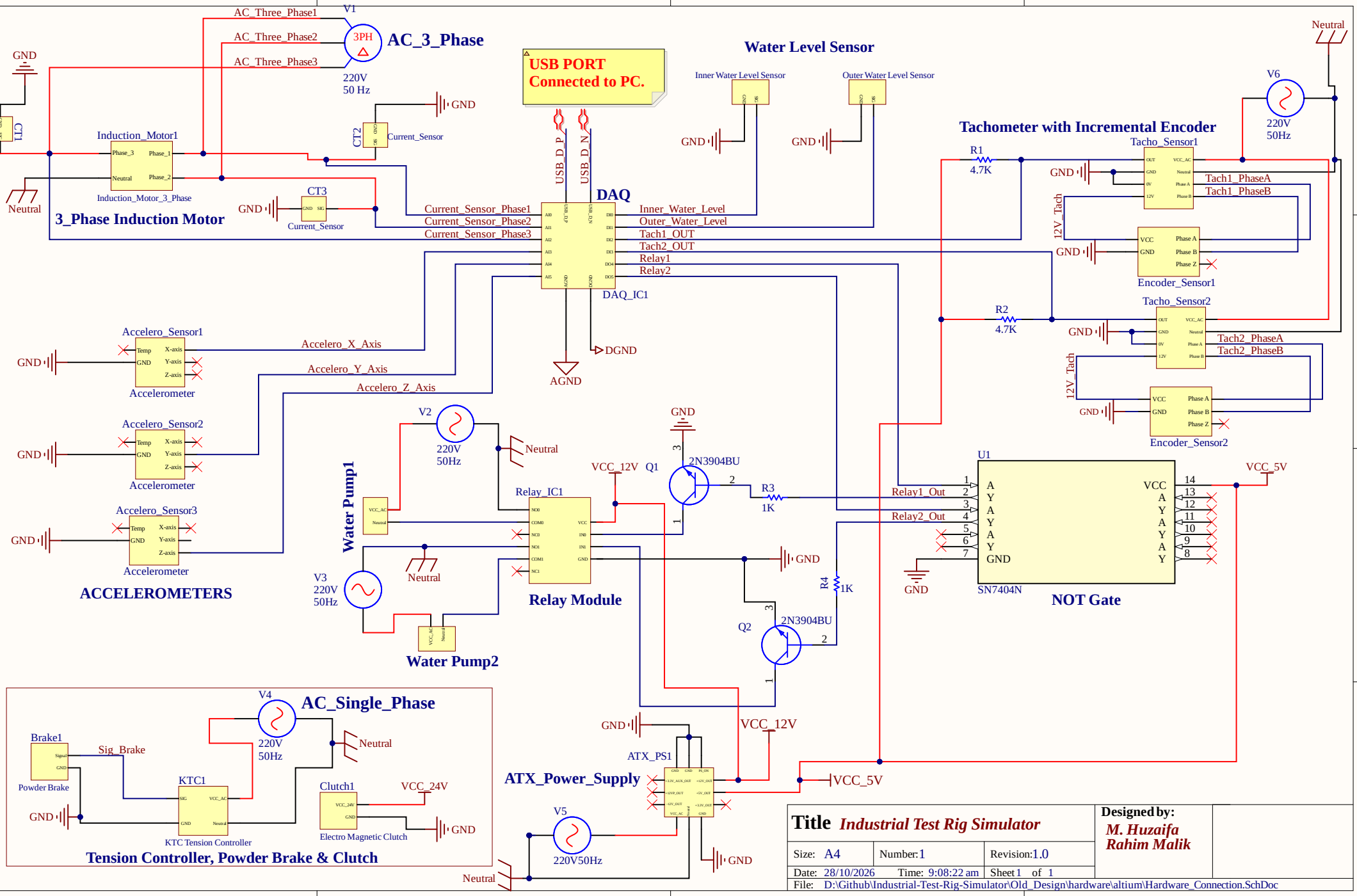

AC_Three_Phase1
AC_Three_Phase2
AC_Three_Phase3
V1
3PH
AC_3_Phase
220V
50 Hz
GND
Current_Sensor
CT1
Induction_Motor1
Phase_3
Phase_1
Neutral
Phase_2
Induction_Motor_3_Phase
3_Phase Induction Motor
CT2
CT3
USB PORT
Connected to PC.
USB_D_P
USB_D_N
DAQ
DAQ_IC1
Current_Sensor_Phase1
Current_Sensor_Phase2
Current_Sensor_Phase3
Inner_Water_Level
Outer_Water_Level
Tach1_OUT
Tach2_OUT
Relay1
Relay2
DGND
AGND
Water Level Sensor
Inner Water Level Sensor
Outer Water Level Sensor
Tachometer with Incremental Encoder
Tacho_Sensor1
Tach1_PhaseA
Tach1_PhaseB
12V_Tach
Encoder_Sensor1
Tacho_Sensor2
Tach2_PhaseA
Tach2_PhaseB
Encoder_Sensor2
R1
4.7K
R2
4.7K
V6
220V
50Hz
Accelero_Sensor1
Accelero_Sensor2
Accelero_Sensor3
Accelerometer
Accelero_X_Axis
Accelero_Y_Axis
Accelero_Z_Axis
ACCELEROMETERS
V2
220V
50Hz
Neutral
Water Pump1
Relay_IC1
VCC_12V
Q1
2N3904BU
R3
1K
Relay Module
V3
220V
50Hz
Water Pump2
Q2
R4
1K
U1
Relay1_Out
Relay2_Out
SN7404N
NOT Gate
VCC_5V
ATX_PS1
ATX_Power_Supply
V5
220V50Hz
V4
AC_Single_Phase
220V
50Hz
Brake1
Sig_Brake
Powder Brake
KTC1
KTC Tension Controller
Clutch1
VCC_24V
Electro Magnetic Clutch
Tension Controller, Powder Brake & Clutch
Title
Industrial Test Rig Simulator
Size: A4
Number: 1
Revision: 1.0
Date: 28/10/2026
Time: 9:08:22 am
Sheet 1 of 1
File: D:\Github\Industrial-Test-Rig-Simulator\Old_Design\hardware\altium\Hardware_Connection.SchDoc
Designed by:
M. Huzaifa
Rahim Malik

Comment	Description	Designator	Footprint	LibRef	Quantity
Accelerometer		Accelero_Sensor1, Accelero_Sensor2, Accelero_Sensor3		Accelerometer	3
ATX_Power_Supply		ATX_PS1		ATX_Supply	1
Powder Brake		Brake1		Powder Brake	1
Electro Magnetic Clutch		Clutch1		Electro Magneetic Clutch	1
Current_Sensor		CT1, CT2, CT3		Current_Sensor	3
NI USB-6008		DAQ_IC1		NI USB-6008	1
Incremental Encoder		Encoder_Sensor1, Encoder_Sensor2		Incremental Encoder	2
Induction_Motor_3_Phase		Induction_Motor1		Induction_Motor_3_Phase	1
Inner Water Level Sensor		Inner_Water_Level_Sensor1		Water Level Sensor	1
KTC Tension Controller		KTC1		KTC Tension Controller	1
Outer Water Level Sensor		Outer_Water_Level_Sensor1		Water Level Sensor	1
2N3904BU	Transistor BJT NPN	Q1, Q2	LM431ACZ	2N3904BU	2
Resistor	Resistor	R1, R2, R3, R4		Resistor	4
Relay		Relay_IC1		Relay	1
Tachometer		Tacho_Sensor1, Tacho_Sensor2		Tachometer	2
SN7404N	No Description Available	U1	N14	SN7404N	1
3-Phase Delta Voltage Source	3-Phase Delta Voltage Source	V1		3 Phase Delta	1
AC Voltage Source	AC Voltage Source	V2, V3, V4, V5, V6		AC Voltage Source	5
Water Pump		Water_Pump1, Water_Pump2		Water Pump	2